Adaptive Context-Guided Segment Aggregation with Low-Drop Consistency for Breast Ultrasound Video Classification

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***Abstract*—Breast ultrasound videos capture a lesion from multiple cross-sections, so video-level classification must aggregate evidence across temporal segments. However, not all segments provide equally reliable diagnostic information. Uniform mean pooling offers a simple and stable aggregation baseline, yet it can dilute discriminative segments; highly selective aggregation, in contrast, may overemphasize isolated local evidence. We propose an adaptive context-guided segment aggregation framework for breast ultrasound video classification. The framework contains two components: adaptive context-guided segment trust gating (ACG-STG) and adaptive context-guided segment perturbation consistency (ACG-SPC). ACG-STG learns residual segment weights from contextual support and prediction-reliability cues while preserving a route close to uniform aggregation. During training, ACG-SPC removes the lowest-weight segment, renormalizes the remaining weights, and encourages the perturbed prediction to remain consistent with the full-video prediction. In inference, only the full adaptive aggregation path is used. The proposed model achieves the highest mean AUC and AUPRC among the compared methods, reaching 72.88%/84.37% and 82.88%/80.41% on two public breast ultrasound video datasets. It also uses fewer parameters and lower measured per-video inference latency than the compared video architectures. Ablation studies show that ACG-STG provides the main improvement, while ACG-SPC gives a small additional gain over matched no-SPC controls when paired with learned segment weights. The code is available at: https://github.com/xuliu9160-ops/ACG-SPC.**

***Index Terms*—breast ultrasound video, video classification, segment aggregation, consistency regularization, multiple instance learning**

I. Introduction

Ultrasound is widely used to assess breast lesions because it supports real-time examination from multiple orientations without ionizing radiation. In routine scanning, a sonographer moves the probe around a lesion and observes changes in its appearance across views. A recorded video, therefore, contains complementary spatial and temporal information that is unavailable in a single manually selected image [1], [2]. However, breast ultrasound interpretation remains challenging because frame quality and lesion appearance across the video can be affected by operator-dependent scanning, speckle noise, acoustic shadowing, and variable lesion morphology [3].

Deep learning has been widely explored for breast ultrasound computer-aided diagnosis, especially for classification from static ultrasound images or representative frames [4]–[7]. Although image-level classifiers can learn discriminative lesion patterns from selected frames, they do not explicitly model how lesion appearance and class-discriminative evidence change across adjacent views during a scan. Recent breast ultrasound video studies have shown that dynamic information can support benign–malignant classification beyond a single selected image [2], [8], [9]. Nevertheless, automated video analysis remains difficult: lesion appearance changes with probe position, only part of a scan may show a discriminative cross-section, and motion or poor acoustic windows can produce weak or unreliable segments [10], [11].

Existing breast ultrasound video methods address temporal heterogeneity by identifying or emphasizing informative frames and segments. For example, KGA-Net uses feature centers derived from static images to guide frame attention [10], while SAG-Net learns segment-level attention weights to emphasize stable video segments [11]. More broadly, this setting is related to attention-based multiple-instance learning and adaptive frame selection, where different instances or frames are allowed to contribute unequally to a video-level prediction [12], [13]. These studies suggest that not all parts of a video should contribute equally. However, they also reveal an important aggregation trade-off. Uniform averaging is stable against occasional extreme segment scores, but it can dilute localized discriminative evidence. Highly selective aggregation can emphasize a salient cross-section, but it may become unstable when the selected evidence is isolated or weakly supported by the rest of the scan.

Figure 1 illustrates the aggregation problem that motivates our design. As shown in Figure 1(a), lesion appearance varies across temporal segments of the same ultrasound video, and some segments provide clearer features for distinguishing benign from malignant lesions than others. This heterogeneity creates the aggregation trade-off in Figure 1(b): uniform mean pooling is stable but may dilute informative segments, whereas highly selective aggregation may over-rely on an isolated high-

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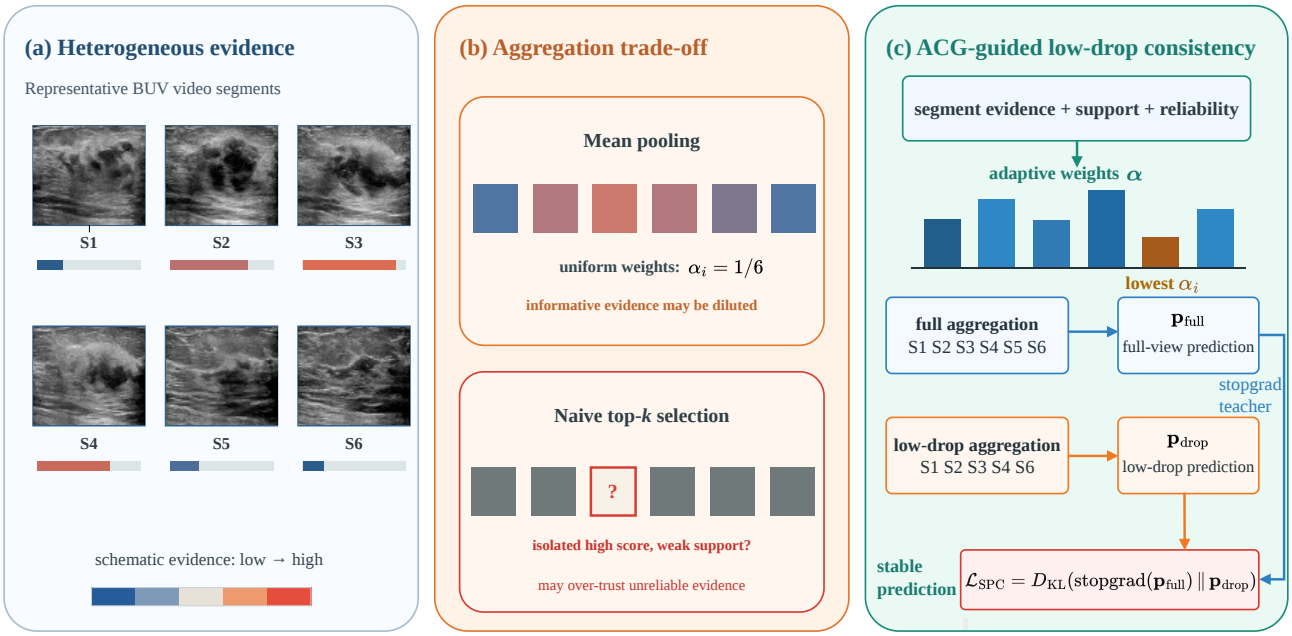


Fig. 1. Motivation for adaptive context-guided aggregation and low-drop consistency. (a) Breast ultrasound video segments vary in lesion appearance and in how clearly they present features useful for benign–malignant classification; the colored bars schematically illustrate the relative importance of different segments for classification. (b) Mean pooling may dilute informative segments, whereas naive top-k selection may over-rely on an isolated segment with weak contextual support. (c) Our framework learns support- and reliability-aware segment weights and uses the lowest-weight segment to define a training-only low-drop consistency perturbation.

score segment that is weakly supported by the remaining video. We therefore seek adaptive aggregation that can emphasize informative segments while retaining the stability of mean pooling. As illustrated in Figure 1(c), the learned segment weights also provide a natural training-time perturbation by removing the lowest-weight segment and encouraging the resulting prediction to remain consistent with the full-video prediction.

To this end, we propose an adaptive context-guided segment aggregation framework with two components. A shared EfficientNet-B0 [14] extracts frame representations, which are aggregated at the segment and video levels. ACG-STG learns residual segment weights from contextual support and prediction-reliability cues, allowing the model to emphasize informative segments while retaining a route close to mean pooling. ACG-SPC further uses the learned weights to define a low-drop consistency objective: during training, the lowest-weight segment is removed, the remaining weights are renormalized, and the perturbed prediction is encouraged to match the full-video prediction. At inference time, only the full adaptive aggregation path is used.

Our evaluation emphasizes protocol-consistent comparisons rather than a cross-paper state-of-the-art claim. We use a fixed evaluation subset with five repeated stratified training–validation splits and report both AUC and AUPRC. Under this protocol, the proposed model achieves the highest mean AUC and AUPRC on both BUV and WHBUS among the compared methods, while requiring fewer parameters and lower measured per-video inference latency than the compared video architectures. The contributions of this work are threefold:

- We introduce ACG-STG, a residual context-guided segment weighting module that adapts video aggregation using support and reliability cues without imposing hard segment selection.
- We formulate ACG-SPC, a weight-guided low-drop consistency objective that regularizes video prediction by removing the lowest-weight segment during training.
- We provide protocol-consistent five-repeat evaluations on BUV and WHBUS, including matched and negative controls that separate the effects of adaptive weighting and low-drop consistency.

II. Related Work

A. Breast Ultrasound Video Analysis

Early deep-learning studies of breast ultrasound concentrated primarily on static images or representative frames, where CNN-based classifiers were trained to distinguish benign and malignant lesions [4]–[7]. Compared with static-image settings, dynamic videos preserve how a lesion and its surrounding tissue appear across probe positions. Zhao *et al.* compared image- and video-based classifiers and showed the utility of dynamic information in a clinical cohort [2]. Prospective and clinically informed systems have subsequently combined videos with BI-RADS descriptors or lesion localization [8], [9]. These works motivate video-level analysis, but their private cohorts and clinical protocols are not directly comparable with the small public benchmarks considered in this study. The public BUV resource was introduced together

with a clip- and video-level aggregation model for lesion detection [1]. For video classification, KGA-Net uses feature centers obtained from a static image dataset to guide frame attention [10], while SAG-Net learns segment-level attention from low-level spatiotemporal features and introduces WH-BUS [11]. Our work shares the goal of learning from multiple video segments, but requires neither an auxiliary static-image dataset nor explicit keyframe selection. It instead focuses on a residual departure from uniform aggregation and uses the learned weights to define a low-drop consistency constraint.

B. Video and Multiple-Instance Aggregation

General video recognition methods often aggregate information from sampled frames or temporal segments. Temporal Segment Networks use sparse temporal sampling and segmental consensus for video-level recognition [15]. R(2+1)D decomposes a 3-D convolution into spatial and temporal operations, while SlowFast uses pathways with different temporal rates [16], [17]. These architectures are useful reference points, but breast ultrasound video datasets are much smaller than common action-recognition corpora. Video-level labels also make segment aggregation related to multiple-instance learning (MIL), where a bag label supervises a set of instances [18]. Attention MIL replaces fixed pooling with a trainable permutation-invariant weighted average [12]. Adaptive frame-selection methods such as AdaFrame learn which observations are sufficient for prediction [13]. Our objective is different from computational pruning: all segment representations remain available to the full-view classifier, while learned residual weights modulate their contribution to the video-level prediction. The low-drop operation is used only as a training perturbation.

C. Consistency under Perturbation

Consistency learning encourages predictions to remain stable under meaning-preserving perturbations of the same sample. Foundational methods such as Mean Teacher use a teacher prediction to regularize the model under perturbed views [19]. In medical image analysis, consistency objectives have also been combined with uncertainty estimation, data augmentation, or prototype learning [20]. Randomly suppressing a segment can be problematic for short medical videos because the removed segment may contain the clearest lesion view. ACG-SPC instead derives the perturbation from the model's learned aggregation: it removes only the lowest-weight segment and renormalizes the remaining weights. The negative mean-pooling control tests whether this weight guidance, rather than consistency alone, accounts for the observed behavior.

III. Method

A. Problem Formulation and Baseline

Let a breast ultrasound video be divided into T temporal segments, each represented by F sampled frames. In all reported experiments, $T = 6$ and $F = 3$. Let $\mathbf{x}_{i,f}$ denote the fth sampled frame from segment i, where $i = 1, \ldots, T$ and $f = 1, \ldots, F$. A shared EfficientNet-B0 encoder ϕ_θ maps each frame to a feature vector,

$$\mathbf{z}_{i,f} = \phi_\theta(\mathbf{x}_{i,f}) \in \mathbb{R}^d. \tag{1}$$

The segment representation is obtained by averaging the frame features within the segment:

$$\mathbf{h}_i = \frac{1}{F} \sum_{f=1}^{F} \mathbf{z}_{i,f}. \tag{2}$$

Let C_ω denote the video-level classifier that maps a video representation to two-class logits. The corresponding predictive distribution is obtained by softmax. The mean-pooling baseline uses

$$\mathbf{h}_{\text{mean}} = \frac{1}{T} \sum_{i=1}^{T} \mathbf{h}_i, \qquad \mathbf{p}_{\text{mean}} = \text{softmax}\left(C_\omega(\mathbf{h}_{\text{mean}})\right). \tag{3}$$

Training uses video-level cross-entropy only:

$$\mathcal{L}_{\text{CE}}(\mathbf{p}, \mathbf{y}) = -\sum_{c=0}^{1} y_c \log p_c, \tag{4}$$

where $\mathbf{y}$ is the one-hot video label and $\mathbf{p}$ is the video-level predictive distribution. For the mean-pooling baseline, $\mathbf{p} = \mathbf{p}_{\text{mean}}$; for the proposed model, $\mathbf{p} = \mathbf{p}_{\text{full}}$. Frame-level loss and frame-branch pretraining are disabled.

B. ACG-STG: Adaptive Residual Weighting

Figure 2 summarizes the complete framework. ACG-STG replaces uniform segment averaging with normalized, sample-dependent weights while retaining an explicit residual route to mean pooling. For each segment, contextual support is computed from the other segment representations:

$$s_i = \frac{1}{T-1} \sum_{j \neq i} \frac{\mathbf{h}_i^\top \mathbf{h}_j}{(\|\mathbf{h}_i\|_2 + \epsilon)(\|\mathbf{h}_j\|_2 + \epsilon)}, \tag{5}$$

where ϵ is a small numerical constant. Let μ_s and σ_s denote the mean and standard deviation of $\{s_i\}_{i=1}^{T}$ within the same video. The standardized support score is

$$\widehat{s}_i = \frac{s_i - \mu_s}{\sigma_s + \epsilon}. \tag{6}$$

The support score is then transformed into a positive trust value:

$$q_i = \sigma\left(\frac{\widehat{s}_i - \tau}{\eta}\right), \qquad t_i = t_{\min} + q_i^\gamma, \tag{7}$$

where $\sigma(\cdot)$ is the sigmoid function, τ is a support threshold, η is a temperature parameter, $t_{\min}$ is a minimum trust floor, and γ controls the sharpness of the trust transformation.

The same classifier head is applied to each segment representation to obtain a segment-level malignant probability:

$$\pi_i = \left[\text{softmax}\left(C_\omega(\mathbf{h}_i)\right)\right]_1, \tag{8}$$

where class 1 denotes malignant. Let $m = \text{median}_i(\pi_i)$. The prediction-reliability term is

$$r_i = \exp(-a|\pi_i - m|), \tag{9}$$

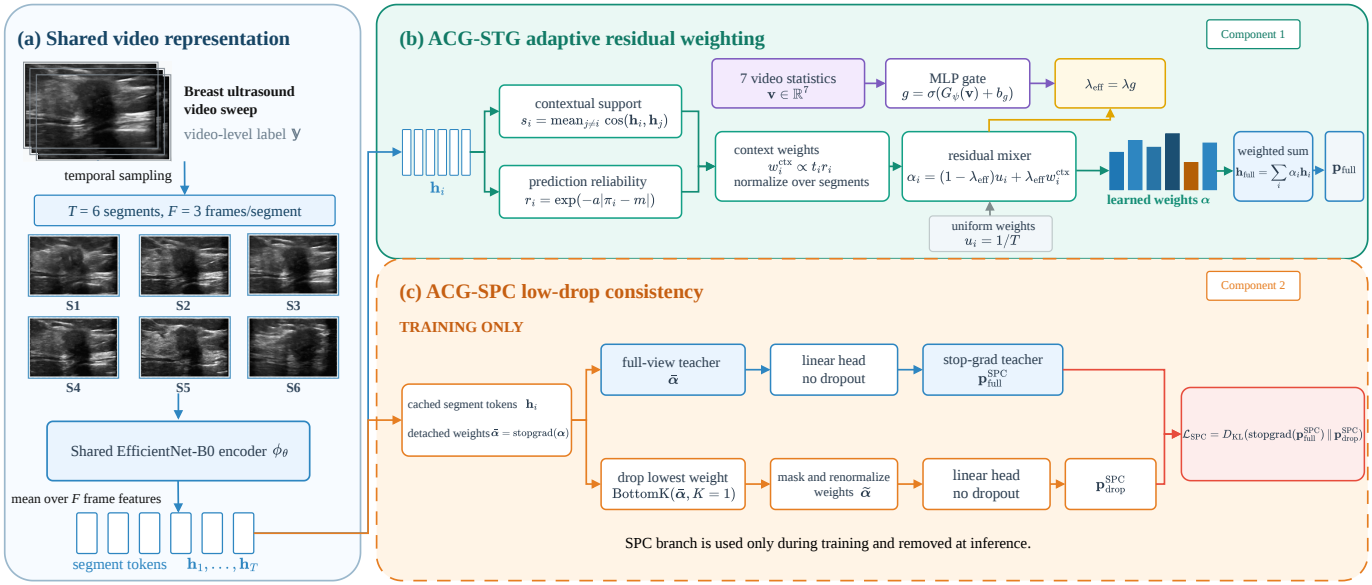


Fig. 2. Overview of the proposed framework. (a) A breast ultrasound video is sampled into six temporal segments with three frames per segment, and a shared EfficientNet-B0 encoder produces segment representations. (b) ACG-STG estimates residual segment weights from contextual support, prediction reliability, and video-level statistics while preserving a route close to uniform pooling. (c) During training, ACG-SPC constructs a full-view teacher path and a low-drop student path using detached aggregation weights. The student path drops the lowest-weight segment, renormalizes the remaining weights, and is coupled to the detached full-view prediction by directed KL consistency. The low-drop branch is removed at inference.

where a controls how strongly isolated segment predictions are downweighted. The normalized context-guided weights are

$$w_i^{\text{ctx}} = \frac{t_i r_i}{\sum_{j=1}^{T} t_j r_j}. \tag{10}$$

The adaptive gate controls how far the model departs from uniform pooling. Let $\mathbf{v} \in \mathbb{R}^7$ denote the detached video-level statistics used by the gate:

$$\mathbf{v} = \text{stopgrad}\left([v_1, \ldots, v_7]\right). \tag{11}$$

The seven statistics summarize the dispersion of segment probabilities, the gap between the highest and average segment probability, the probability range, adjacent-segment prediction variation, mean contextual support, the support of the highest-probability segment, and normalized probability concentration. A two-layer MLP produces a scalar gate:

$$g = \sigma(G_\psi(\mathbf{v}) + b_g), \qquad \lambda_{\text{eff}} = \lambda g, \tag{12}$$

where b_g is the gate bias and $\lambda \in [0, 1]$ is the maximum residual-mixing coefficient. Therefore, $\lambda_{\text{eff}} \in [0, 1]$.

Let $\mathbf{u} = (1/T, \ldots, 1/T)$ denote uniform segment weights. The final adaptive weights are

$$\alpha_i = (1 - \lambda_{\text{eff}})u_i + \lambda_{\text{eff}} w_i^{\text{ctx}}, \qquad \sum_{i=1}^{T} \alpha_i = 1. \tag{13}$$

The full adaptive video representation and prediction are

$$\mathbf{h}_{\text{full}} = \sum_{i=1}^{T} \alpha_i \mathbf{h}_i, \qquad \mathbf{p}_{\text{full}} = \text{softmax}\left(C_\omega(\mathbf{h}_{\text{full}})\right). \tag{14}$$

This residual design allows adaptive segment emphasis without assuming that a single segment is sufficient for the video-level classification decision.

C. ACG-SPC: Low-Drop Consistency

ACG-SPC uses the learned weights to construct a second aggregation view during training. The learned weights are detached before selecting the removed segment:

$$\bar{\boldsymbol{\alpha}} = \text{stopgrad}(\boldsymbol{\alpha}). \tag{15}$$

Only the aggregation weights are detached; the segment representations remain in the computation graph for the low-drop student path. Let

$$\mathcal{M}_{\text{low}} = \text{BottomK}\left(\{\bar{\alpha}_i\}_{i=1}^{T}, K\right) \tag{16}$$

be the indices of the K lowest-weight segments. We use $K = 1$ in all experiments. The selected segment is masked in aggregation space, and the remaining weights are renormalized:

$$\tilde{\alpha}_i = \frac{\bar{\alpha}_i \mathbb{I}[i \notin \mathcal{M}_{\text{low}}]}{\sum_{j=1}^{T} \bar{\alpha}_j \mathbb{I}[j \notin \mathcal{M}_{\text{low}}] + \epsilon}. \tag{17}$$

The ϵ term is used only for numerical stability.

Both SPC paths use the linear classifier head with dropout disabled, avoiding additional stochasticity in the consistency branch. Denote this deterministic classifier by C_ω^{det}. The full-view teacher distribution and low-drop student distribution are

$$\begin{aligned} \mathbf{p}_{\text{full}}^{\text{SPC}} &= \text{softmax}\left(C_\omega^{\text{det}}\left(\sum_{i=1}^{T} \bar{\alpha}_i \mathbf{h}_i\right)\right), \\ \mathbf{p}_{\text{drop}}^{\text{SPC}} &= \text{softmax}\left(C_\omega^{\text{det}}\left(\sum_{i=1}^{T} \tilde{\alpha}_i \mathbf{h}_i\right)\right). \end{aligned} \tag{18}$$

If ACG-STG assigns little contribution to the removed segment, the two video predictions should remain close. We impose a directed consistency objective using the detached full-view prediction as the teacher:

$$\mathcal{L}_{\mathrm{SPC}} = D_{\mathrm{KL}}\left(\mathrm{stopgrad}\left(\mathbf{p}_{\mathrm{full}}^{\mathrm{SPC}}\right) \,\|\, \mathbf{p}_{\mathrm{drop}}^{\mathrm{SPC}}\right). \tag{19}$$

The full training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}}(\mathbf{p}_{\mathrm{full}}, \mathbf{y}) + \beta_{\mathrm{SPC}}(e)\mathcal{L}_{\mathrm{SPC}}, \tag{20}$$

where e denotes the training epoch. The SPC coefficient is linearly ramped during the first five epochs:

$$\beta_{\mathrm{SPC}}(e) = \beta_{\mathrm{SPC}}^{\max} \min\left(\frac{e}{5}, 1\right). \tag{21}$$

The low-drop branch is discarded at inference, so prediction uses only $\mathbf{p}_{\mathrm{full}}$ from the full adaptive representation.

IV. Experiments

A. Datasets

We evaluate binary breast ultrasound video classification on BUV and WHBUS, the two public datasets used by the closely related SAG-Net study [11]. Following the classification manifests used in all reported experiments, BUV contains 185 videos, including 71 benign and 114 malignant videos, and WHBUS contains 184 videos, including 101 benign and 83 malignant videos. BUV originates from a public breast ultrasound video resource introduced for lesion detection [1], whereas WHBUS was collected for video-level classification [11]. For each dataset, we follow the official train/validation split. The official validation split is used as a fixed evaluation set and is not used for model selection. The official training split forms the development pool, which is stratified into model-fitting and internal-validation subsets in each of five repeats. The internal-validation subset is used for early stopping, checkpoint selection, threshold selection, and development-time coefficient selection, while the fixed evaluation set is evaluated once per repeat.

B. Repeated Evaluation Protocol

For each dataset, a fixed evaluation subset is held constant across all runs. In each of five repeats, the remaining training pool is split into 85% model fitting and 15% internal validation using stratified sampling to preserve the benign–malignant proportion. Early stopping uses internal-validation AUC with patience 5, and the checkpoint with the best internal-validation AUC is evaluated once on the fixed evaluation subset. Final results are reported as mean and standard deviation over the five repeats. This repeated design reduces dependence on a single training–validation split while keeping the evaluation subset unchanged across repeats.

C. Implementation and Compared Methods

All controlled EfficientNet-B0 experiments use six temporal segments with three consecutive frames per segment, input size 224×224, batch size 8, a maximum of 50 epochs, five repeats, and base seed 42. Training samples one anchor within each temporal segment and uses three consecutive frames; validation and testing use deterministic center sampling. Training augmentation applies clip-consistent vertical flipping and small rotations from $\{-15, -10, -5, 5, 10, 15\}^{\circ}$. Optimization uses AdamW with learning rate 10^{-4}, weight decay 10^{-3}, cosine annealing to 10^{-6}, gradient clipping at 1.0, and ImageNet initialization. Frame-level loss, frame-branch pretraining, and rank loss are disabled. The mean baseline averages the six segment representations and optimizes video-level cross-entropy.

ACG-STG uses the same fixed support and reliability hyperparameters for all datasets: $\tau = 0$, $\eta = 1$, $t_{\min} = 0.05$, $\gamma = 1$, $a = 1$, and a median-based reliability center. We restrict dataset-specific development to three coefficients: the maximum residual-mixing coefficient λ, the gate bias b_g, and the maximum SPC consistency coefficient $\beta_{\mathrm{SPC}}^{\max}$. These coefficients are selected using the internal-validation procedure within the development pool and are not selected on the fixed evaluation set. For the ACG-STG-only configuration, BUV uses $\lambda = 0.20$ and $b_g = -0.5$, while WHBUS uses $\lambda = 0.25$ and $b_g = -0.25$. For the final ACG-STG + ACG-SPC model, BUV uses $\lambda = 0.15$, $b_g = 0$, and $\beta_{\mathrm{SPC}}^{\max} = 0.01$; WHBUS uses $\lambda = 0.25$, $b_g = -0.25$, and $\beta_{\mathrm{SPC}}^{\max} = 0.00625$. Both datasets use $K = 1$ low-drop segment and a five-epoch SPC warmup.

For R(2+1)D-18 [16], SlowFast R50 [17], and SAG-Net [11], we use their default initialization and the same five-repeat data-split/evaluation protocol. EfficientNet-B0 Mean and our variants additionally share the backbone, sampling pattern, batch size, training duration, and checkpoint-selection rule. The implementation uses Python 3.8.10, PyTorch 1.12.1, torchvision 0.13.1, timm 0.9.12, NumPy 1.24.4, and scikit-learn 1.3.2. Per-video latency is measured with batch size 1 on NVIDIA GeForce RTX 3090, excluding data loading.

D. Evaluation Metrics

AUC and AUPRC are the primary metrics and are reported as percentages, with malignant as the positive class. AUC summarizes ranking across false-positive rates, whereas AUPRC summarizes the precision–recall trade-off for the malignant class.

V. Results

A. Main Comparisons

Table I summarizes the five-repeat AUC/AUPRC results together with model size and measured per-video inference time. Under the common protocol, Ours achieves the highest mean AUC and AUPRC on both datasets, reaching $72.88/84.37$ on BUV and $82.88/80.41$ on WHBUS, while using 5.65M parameters and 7.828 ms per video. ACG-SPC adds no inference branch because the low-drop path is used only during training.

B. Ablation Analysis

Table II separates adaptive weighting from low-drop consistency. On BUV, STG-only improves AUC/AUPRC over mean pooling by 0.64/1.05 points; under the matched final coefficient setting, SPC further changes them from 72.69/84.28

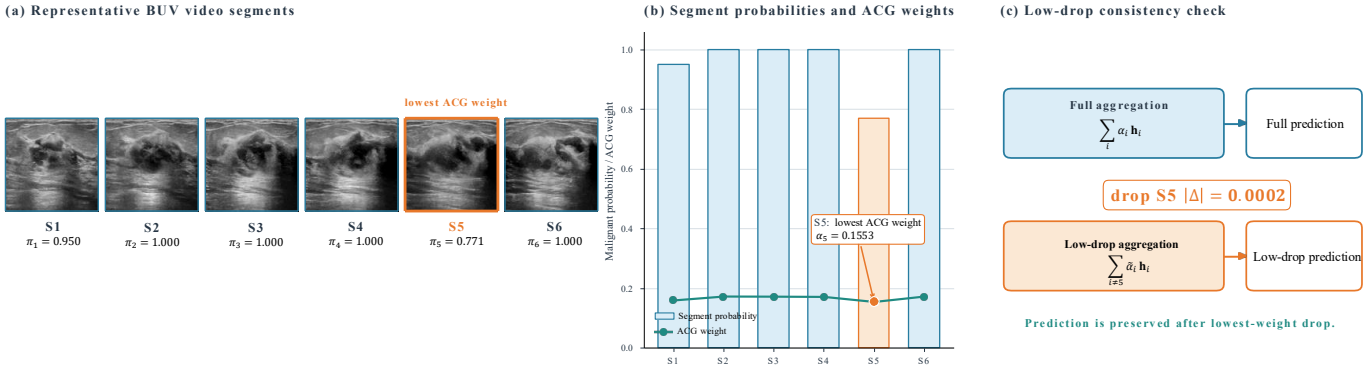


Fig. 3. Example final-model output for a malignant BUV video. (a) Representative frames from the six temporal segments, with segment malignant probabilities π_i shown below each frame. (b) Segment malignant probabilities and normalized ACG weights; S5 receives the lowest ACG weight. (c) Low-drop consistency check: S5 is removed and the remaining weights are renormalized, yielding a small video-level malignant-probability change ($|\Delta| = 0.0002$). This single-case visualization illustrates the low-drop perturbation mechanism.

TABLE I
MAIN VIDEO-LEVEL CLASSIFICATION RESULTS AND INFERENCE EFFICIENCY.

Method	BUV AUC / AUPRC	WHBUS AUC / AUPRC	Efficiency Params / Time
R(2+1)D-18 [16]	70.32 ± 1.18 78.42 ± 2.59	77.29 ± 7.48 77.71 ± 5.72	31.30M 29.422 ms
SlowFast R50 [17]	70.77 ± 10.52 80.59 ± 9.46	75.06 ± 6.67 75.88 ± 7.19	33.65M 13.487 ms
SAG-Net [11]	70.71 ± 4.67 80.50 ± 4.03	74.82 ± 2.84 70.68 ± 3.74	27.84M 27.250 ms
Ours	$\mathbf{72.88 \pm 2.78}$ $\mathbf{84.37 \pm 3.07}$	$\mathbf{82.88 \pm 2.46}$ $\mathbf{80.41 \pm 5.00}$	5.65M 7.828 ms

Note: AUC/AUPRC: mean $\pm$ SD (%) over five repeats with repeated-split training and fixed evaluation. Time/video is per video. Ours is ACG-STG+SPC.

TABLE II
CONTROLS FOR ADAPTIVE WEIGHTING AND LOW-DROP CONSISTENCY.

Variant	AUC	AUPRC
BUV		
Mean	72.24 ± 3.06	83.22 ± 2.90
STG-only	72.88 ± 2.69	84.27 ± 2.83
STG-noSPC	72.69 ± 2.84	84.28 ± 3.09
STG+SPC	72.88 ± 2.78	84.37 ± 3.07
Mean+SPC	69.74 ± 2.44	82.30 ± 3.20
WHBUS		
Mean	81.53 ± 5.66	77.65 ± 9.53
STG-only	82.53 ± 3.01	80.18 ± 4.96
STG+SPC	82.88 ± 2.46	80.41 ± 5.00
Mean+SPC	81.41 ± 3.44	79.42 ± 4.95

Note: Mean $\pm$ SD (%). STG = ACG-STG. STG-only uses coefficients selected without SPC; STG-noSPC uses final STG+SPC coefficients with SPC disabled (identical on WHBUS, so omitted). Mean+SPC applies SPC to uniform pooling.

to 72.88/84.37. On WHBUS, STG-only improves over mean pooling by 1.00/2.53 points, and SPC adds 0.35/0.23 points. Mean+SPC does not provide a consistent benefit and remains below the final model, supporting learned weights as the basis for the low-drop perturbation without treating them as clinical explanations.

Table III examines segment selection, residual mixing, and context cues. Top-k softmax is weaker on BUV than the

TABLE III
COMPONENT ABLATIONS FOR ADAPTIVE SEGMENT AGGREGATION.

Variant	AUC	AUPRC
BUV		
Top-k softmax	68.97 ± 2.20	81.87 ± 1.37
No residual	72.56 ± 2.00	85.02 ± 1.60
No support	72.50 ± 2.61	84.18 ± 2.92
No reliability	72.88 ± 2.79	84.35 ± 2.83
STG-only	72.88 ± 2.69	84.27 ± 2.83
STG+SPC	72.88 ± 2.78	84.37 ± 3.07
WHBUS		
No residual	82.06 ± 3.15	79.08 ± 4.80
No support	82.35 ± 2.13	79.54 ± 4.59
No reliability	82.35 ± 3.08	80.31 ± 5.49
STG-only	82.53 ± 3.01	80.18 ± 4.96
STG+SPC	82.88 ± 2.46	80.41 ± 5.00

Note: Mean $\pm$ SD (%). STG = ACG-STG. "No residual" removes residual mixing; "No support"/"No reliability" remove the corresponding cue. Top-k softmax is BUV-only.

residual adaptive variants. Removing the support cue lowers both metrics relative to the final model on both datasets, while removing the reliability cue produces a smaller change. The no-residual variant is mixed on BUV but lower than the final model on WHBUS, suggesting that the value of residual mixing is dataset dependent.

VI. DISCUSSION

Controlled comparisons: The experiments identify adaptive weighting as the main source of improvement. This is consistent with ultrasound sweeps, where some segments may provide clearer morphology or stronger context while hard selection remains risky; residual weighting offers a middle ground between uniform pooling and sparse selection. ACG-SPC has a smaller complementary role by encouraging stability after removal of the least-used segment. The Mean+SPC control shows that consistency alone is insufficient: with uniform weights, "lowest-weight" is arbitrary and performance does not improve consistently. This supports learned weights for defining the perturbation, not as validated segment-level clinical explanations.

Relation to clinical video analysis: Breast ultrasound video analysis is attractive because a scan records multiple views without requiring a radiologist to select a single image in advance [2], [11]. Our framework operationalizes the idea that recorded moments should not contribute equally at the segment-aggregation level, without requiring keyframe labels or an auxiliary static-image dataset. The learned weights may help inspect model behavior, but they are not validated explanations or lesion localizations. Clinical usefulness would require patient-level external validation, prospective assessment, and failure-case analysis.

Limitations: The proposed framework has several limitations. First, segment weights in ACG-STG are learned only from video-level labels, without segment-level lesion annotations, keyframe labels, or expert importance scores. Therefore, a high weight should be interpreted as a model-dependent contribution to aggregation, not as verified diagnostic evidence or lesion localization. When a video contains multiple complementary views, the model may emphasize classifier-friendly segments while underweighting other clinically informative views.

Second, ACG-STG uses contextual support and prediction reliability as simplified summaries of an ultrasound sweep. Diagnostic evidence may depend on lesion morphology, boundary characteristics, posterior acoustic patterns, probe motion, and surrounding tissue, which are not explicitly disentangled by segment-feature aggregation. The weights may therefore be unstable when features fail to capture lesion structure or when videos contain strong artifacts, abrupt motion, or incomplete lesion views. Evaluation is also limited to two small public datasets, so broader validation and threshold calibration remain necessary.

VII. Conclusion

We presented a context-guided residual segment aggregation framework for breast ultrasound video classification. ACG-STG learns residual segment weights from contextual support and reliability cues, while ACG-SPC uses them to define a low-drop consistency perturbation during training. Under the five-repeat fixed-evaluation protocol, the final model achieves the highest mean AUC and AUPRC on both datasets, with fewer parameters and lower per-video inference latency than the compared video architectures. Ablations identify adaptive weighting as the main contributor, while matched and negative controls show that SPC provides a smaller complementary benefit when paired with learned segment weights rather than uniform mean pooling.

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